

Data Mining I

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INF

FACULTY OF
COMPUTER SCIENCE



- 1 What is Data Mining for?
- 2 Course overview & literature
- 3 Administrative issues

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Diagnosis, eg for NAFLD

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! NAFLD has no symptoms.

! The gold standard for NAFLD diagnosis is liver biopsy.

→ How can we diagnose NAFLD (early) without invasion?

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Treatment planing, eg for chronic innitus

"Tinnitus represents a relatively common condition in the global population accompanied by various comorbidities and severe burden in many cases. Nevertheless, there is currently no general treatment or cure, presumable due to the heterogeneity of tinnitus with its wide variety of etiologies and tinnitus phenotypes. Hence, most treatment studies merely demonstrated improvement in a subgroup of tinnitus patients. . . ." ^a

→ How to select the right treatment for a patient?

^aSchoisswohl S, Langguth B, Schecklmann M, Bernal-Robledano A, Boecking B, Cederroth CR, Chalanouli D, Cima R, Denys S, Dettling-Papargyris J, Escalera-Balsera A. Unification of Treatments and Interventions for Tinnitus Patients (UNITI): a study protocol for a multi-center randomized clinical trial. Trials. 2021 Dec;22(1):1-6.

How can DM help . . .

choose the right treatment for a patient ?

- Q1: How to predict whether the treatment will bring improvement or not?
- Q4: How to find out which phenotypes benefit / do not benefit from the treatment?
- Q9: Is the finding reliable ??
- Q10: How to select the right treatment for a patient?

DM I: Goal of the course

In this course you learn

- ▶ to understand data mining methods
- ▶ and to use them for simplified but real problems.

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Course contents

Main thread

- ★ **Block 1:** Classification – Underpinnings, Algorithms & Evaluation
- ★ **Block 2:** Clustering – Underpinnings, Algorithms & Evaluation
- ★ **Block 3:** Frequent itemset discovery

Orthogonal issues

- ★ **Block 'Evaluation':** More on evaluation / assessment of model quality
- ★ **Block 'Data':** Data quality & data preprocessing

Further books and papers in the slideset of each lecture

Literature

Book of the course

Pan-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar. *Introduction to Data Mining*, PEARSON, 2019 (2nd edition) ^a

- ▶ Chapters 2, 3
- ▶ Parts of Chapters 4, 6, 7, 8, 10

See overview under

<https://www-users.cs.umn.edu/~kumar001/dmbook/index.php> with links to the book and its electronic version, to sample chapters, and to slide collections (version 2018)

^aSome DM I slides come from the 1st edition. They concern content that did not change between the two editions.

For the Block 'Data'

Salvador Garcia, Julian Luengo, Francisco Herrera *Data Preprocessing in Data Mining*, SPRINGER International Publishing Switzerland, 2015

- ▶ Chapter 3 'Data Preparation Basic Models'
- ▶ Chapter 4 'Dealing with Missing Values'
- ▶ Chapter 7 'Feature Selection'

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Intended participants:

- ▶ FIN students of the 4-semester master degrees
- ▶ Students of other faculties (FIN export)

This course brings 6 ECTS.

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- ▶ Exercise sheets
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Examination

- ▶ Written exam with
- ▶ 'Votierung' as prerequisite → Information in moodle and in the Exercises

Questions thus far?